Cart pole equations of motion

Edgar Jose Donoso Mansilla

1 Variables

Simplifying the problem by treating it as 2 point masses where c is the point on the cart and then p is at the other end of the pole. The masses are at a ditance l of each other.

The problem needs 2 generalized variables:

1. The horizontal displacement of the cart: x
2. The angle to the up axis of the pole: θ

2 Kinematics

Finding the position of the mass at the end of the pole (x_p, y_p):

$$x_p = x - l\sin(\theta) \tag{1}$$

$$y_p = l\cos(\theta) \tag{2}$$

Now finding the first order derivative for the kinetic speed:

$$\dot{x_p} = \dot{x} - l\dot{\theta}\cos(\theta) \tag{3}$$

$$\dot{y_p} = -l\dot{\theta}\sin(\theta) \tag{4}$$

3 Lagrangian

The kinetic energy for the cart is:

$$T = \frac{1}{2}m_c\dot{x}^2 + \frac{1}{2}m_p(\dot{x_p}^2 + \dot{y_p}^2)$$
$$T = \frac{1}{2}m_c\dot{x}^2 + \frac{1}{2}m_p(\dot{x}^2 - 2\dot{x}l\dot{\theta}\cos(\theta) + l^2\dot{\theta}^2\cos^2(\theta) + l^2\dot{\theta}^2\sin^2(\theta))$$

Simplifying with trigonometric identities:

$$T = \frac{1}{2}m_c\dot{x}^2 + \frac{1}{2}m_p(\dot{x}^2 - 2\dot{x}l\dot{\theta}\cos(\theta) + l^2\dot{\theta}^2) \tag{5}$$

The potential energy for the cart is:

$$U = mgy_p$$

$$U = m_p gl\cos(\theta) \tag{6}$$

This leads to the final lagrangian of:

$$L = T - U$$
$$L = \frac{1}{2}m_c\dot{x}^2 + \frac{1}{2}m_p(\dot{x}^2 - 2\dot{x}l\dot{\theta}\cos(\theta) + l^2\dot{\theta}^2) - m_p gl\cos(\theta)$$

Which simplifies to the following:

$$L = \frac{1}{2}(m_c + m_p)\dot{x}^2 - m_p\dot{x}l\dot{\theta}\cos(\theta) + \frac{1}{2}m_p l^2\dot{\theta}^2 - m_p gl\cos(\theta) \tag{7}$$

4 Euler Lagrange Equation

To optimize the energy ledger we have made, we can now use the Euler-Lagrange equation to get the final equation of motion for both x and θ.

With respect to x:

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = \mathbf{F}(t)$$

$$\frac{\partial L}{\partial \dot{x}} = (m_c + m_p)\dot{x} - m_p l\dot{\theta}\cos(\theta) \tag{8}$$
$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = (m_c + m_p)\ddot{x} - m_p l\ddot{\theta}\cos(\theta) + m_p l\dot{\theta}^2\sin(\theta) \tag{9}$$
$$\frac{\partial L}{\partial x} = 0 \tag{10}$$

$$(m_c + m_p)\ddot{x} - m_p l\ddot{\theta}\cos(\theta) + m_p l\dot{\theta}^2\sin(\theta) = \mathbf{F}(t) \tag{11}$$

With respect to θ:

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0$$

$$\frac{\partial L}{\partial \dot{\theta}} = -m_p\dot{x}l\cos(\theta) + m_p l^2\dot{\theta} \tag{12}$$
$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} = -m_p\ddot{x}l\cos(\theta) + m_p\dot{x}l\dot{\theta}\sin(\theta) + m_p l^{2\ddot{}}(\theta) \tag{13}$$
$$\frac{\partial L}{\partial \theta} = m_p\dot{x}l\dot{\theta}\sin(\theta) + m_p gl\sin\theta \tag{14}$$

$$-m_p\ddot{x}l\cos(\theta) + m_p\dot{x}l\dot{\theta}\sin(\theta) + m_p l^2\ddot{(\theta)} + m_p\dot{x}l\dot{\theta}\sin(\theta) + m_p gl\sin\theta = 0 \quad (15)$$

Equation (15) can be simplified by dividing the mass out as it is not being influenced by any forces:

$$-\ddot{x}l\cos(\theta) + \dot{x}l\dot{\theta}\sin(\theta) + l^2\ddot{(\theta)} + \dot{x}l\dot{\theta}\sin(\theta) + gl\sin\theta = 0 \quad (16)$$

5 Summary

To summarize, the equations of motion with respect to displacement x and angle θ are:

$$(m_c + m_p)\ddot{x} - m_p l\ddot{\theta}\cos(\theta) + m_p l\dot{\theta}^2\sin(\theta) = \mathbf{F}(t) \quad (17)$$

$$-\ddot{x}l\cos(\theta) + \dot{x}l\dot{\theta}\sin(\theta) + l^2\ddot{(\theta)} + \dot{x}l\dot{\theta}\sin(\theta) + gl\sin\theta = 0 \quad (18)$$